AI-Assisted Evaluation of the Antibacterial Activity of Methanolic Extract of *Chroococcus* sp. against Selected Bacteria

Scan for the online poster

Kholoud Khalid Alotaibi. ID: 445203007 | Email: 445203007@student.ksu.edu.sa
Supervisor: Dr. Hend Alwathnani | Email: wathnani@ksu.edu.sa
Department of Botany and Microbiology, College of Sciences, King Saud University

Abstract: The growing concern over antibiotic resistance has encouraged research into natural bioactive compounds, particularly those derived from cyanobacteria such as Chroococcus sp. This study evaluated its methanolic extract (0.5–10 mg/mL) against four Gram-positive and Gram-negative bacteria using the agar well diffusion method. Significant antibacterial activity was observed against E. coli at 5 mg/mL (+7.0 mm, p = 0.032) and P. aeruginosa at 10 mg/mL (+12.3 mm, p = 0.006), compared with the methanol control. No significant activity was detected against S. aureus or B. subtilis. MIC determination was unsuccessful due to extract color interference with OD600 measurements. Machine-learning models (ridge regression and random forest) showed limited predictive performance (MAE ≈ 3 mm, $R^2 \approx 0$), with no meaningful improvement over the baseline model.

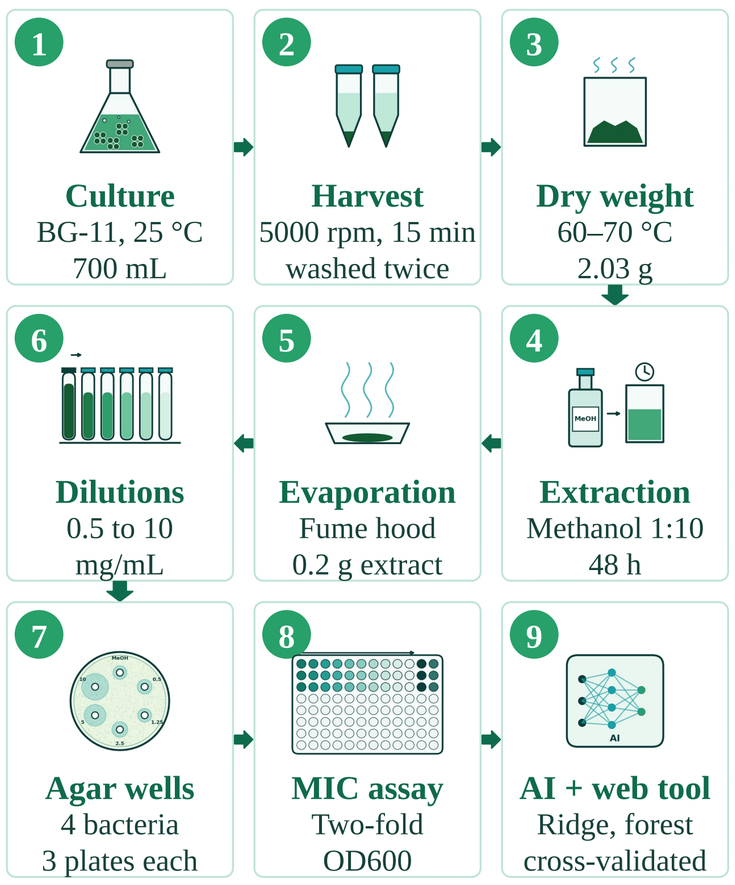
Introduction

Natural products derived from cyanobacteria, particularly *Chroococcus* sp., represent promising sources of antibacterial compounds against Gram-positive and Gram-negative bacteria .Their antibacterial activity varies according to species, growth conditions, extraction methods, and bacterial susceptibility (Gheda & Ismail, 2020; Alsenani et al., 2020). The agar well diffusion method is widely used for preliminary screening, while machine learning can help analyze antibacterial and predict inhibition zones based on extract concentration and bacterial species (Balouiri et al., 2016; Rahman et al., 2022).

Objectives

- Evaluate methanolic extract of *Chroococcus* sp antibacterial.
- Evaluate antibacterial activity at five concentrations by agar well diffusion.
- Determine the minimum inhibitory concentration (MIC).
- Compare the responses of Gram-negative and Gram-positive bacteria.
- Investigate AI models for predicting inhibition zones.

Materials & Method



Workflow: preparation → experiment → measurement → analysis.

Results

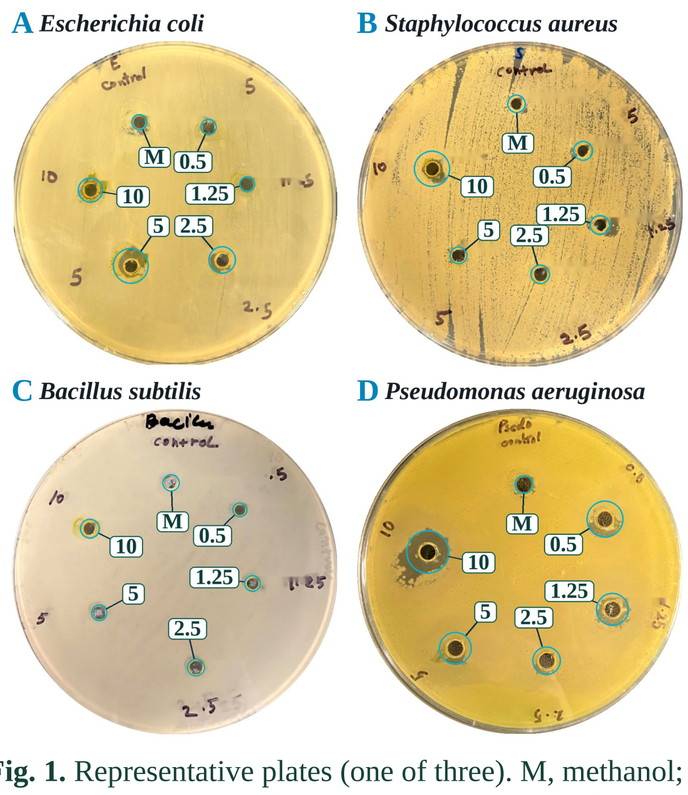


Fig. 1. Representative plates (one of three). M, methanol; 0.5–10, extract (mg/mL); circles outline measured zones.

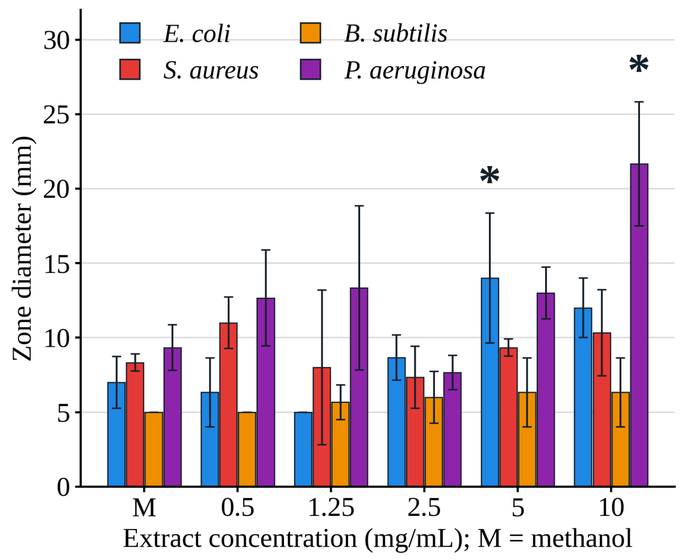


Fig. 2. Zone diameter (mean ± SD, n = 3). * $p < 0.05$ vs methanol (Tukey's test).

Table 1. Tests on the zones of each bacterium.

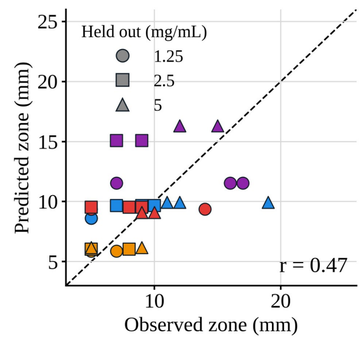
Bacterium	ANOVA p	vs methanol	Slope
E. coli	0.004	5 mg/mL: +7.0 mm (p = 0.032)	1.87 (0.003)
S. aureus	0.55	none	−0.05 (0.92)
B. subtilis	0.78 K–W	none	0.32 (0.25)
P. aeruginosa	0.004	10 mg/mL: +12.3 mm (p = 0.006)	1.58 (0.096)

Tukey's test vs methanol. K–W, Kruskal-Wallis. Slope in mm per doubling of concentration (p).

MIC:

- OD600 followed the extract color, not growth
- No well reached 90% inhibition
- Extract only: 0.64–0.96 vs growth control 0.22–0.48

Results: AI Models



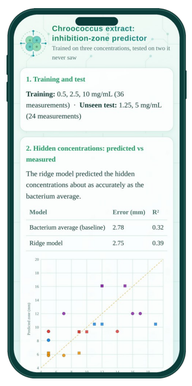


Fig. 3. Observed vs predicted zones (ridge, leave-one-concentration-out): MAE 2.96 mm vs 3.05 mm for the bacterium mean, $R^2 \approx 0$; better only for *E. coli* (2.97 vs 4.12 mm). Right: the interactive web tool.

Scan to open the web tool

Discussion

Agar diffusion results indicated antibacterial activity of the methanolic *Chroococcus* sp. extract against *E. coli* and *P. aeruginosa*, although broth-based results were inconclusive due to OD600 interference. AI models showed limited predictive performance and did not outperform the baseline; therefore, further studies using extract-specific blanks, viable cell counts, additional replicates, and larger datasets are recommended to confirm antibacterial activity and determine MIC.

Conclusions & Recommendations

Chroococcus sp. extract showed selective antibacterial activity against E. coli and P. aeruginosa. Further studies are recommended to confirm MIC using extract-specific blanks and viable bacterial counts.

References

Acknowledgments

Thanks to the Department and to Dr. Hend Alwathnani and the laboratory technicians Ms. Sumaiah Al-Shehri and Ms. Huriyah Al-Dahish.

Copyright © 2026